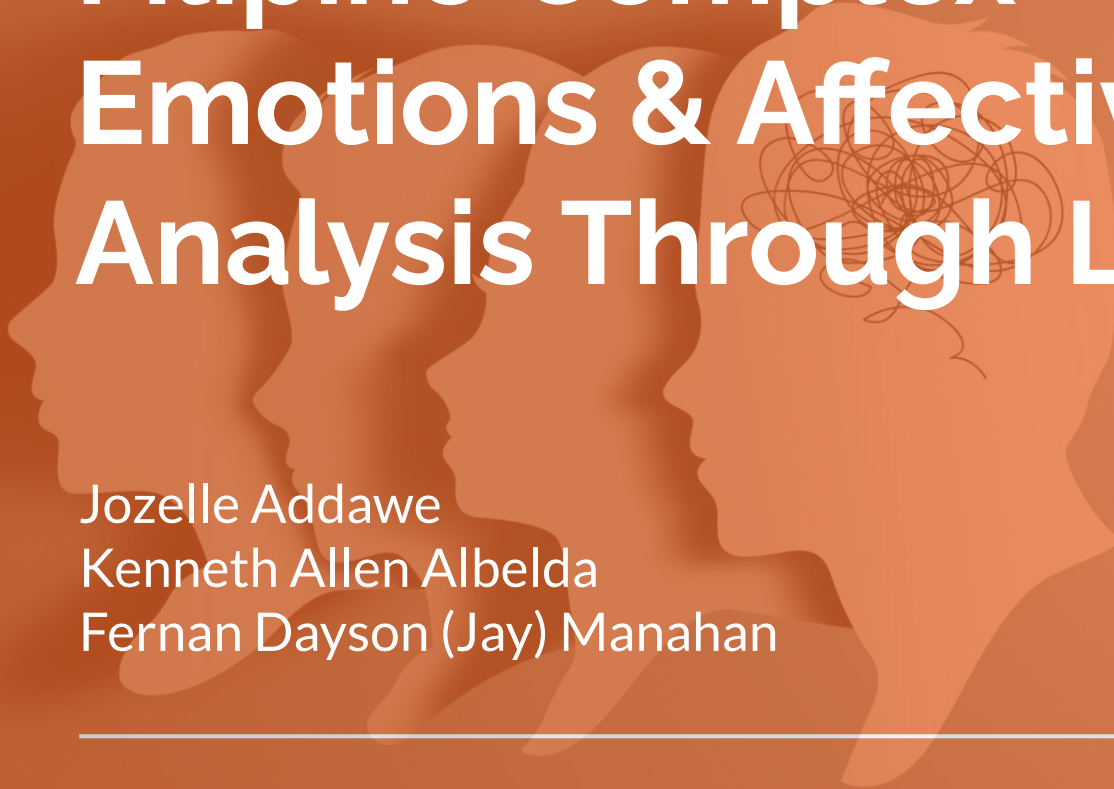

Filipino Complex Emotions & Affective Analysis Through LLMs



Jozelle Addawe
Kenneth Allen Albelda
Fernan Dayson (Jay) Manahan

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Motivation of the Study

Unlike English, Filipino is an under-resourced language in NLP.

One such area where Filipino has a low-resource status is the important field of emotion classification and affective analysis.

We wish to evaluate if LLM-based English emotion classifiers can perform just as well in Filipino, given the language's unique morphological and syntactic structures.

We also build a corpus of Filipino statements that can be used for further LLM-based affective analyses.

We expand on prior works that separately built a Filipino depression corpus (Tumaliuan et al., 2024) and another that benchmarked pre-trained LLMs' performance on the Filipino language (Montalan et al., 2025).

—

How can we design a large
language model to recognize
**complex emotions expressed
in Filipino?**

Related Literature

Pandey (2021): [Emotion Dataset for Emotion Recognition Tasks](#)

Balatska (2024): [Mental Health Sentiment Analysis](#)

Montalan et al. (2025): [Batayan: A Filipino NLP benchmark for evaluating Large Language Models](#)

Tumaliuan et al. (2024): [Development of Depression Data Sets and a Language Model for Depression Detection: Mixed Methods Study](#)

Liu et al. (2019): [ROBERTA: A robustly optimized BERT pretraining approach](#)

Cruz and Cheng (2021): [Improving large-scale language models and resources for Filipino](#)

Conneau et al. (2019): [Unsupervised Cross-lingual Representation Learning at Scale](#)

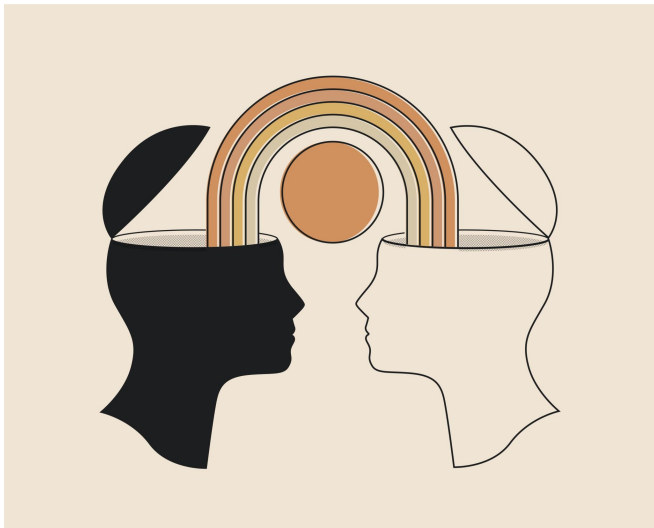
Devlin et al. (2018): [BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding](#)

Related Literature

- **Classification of Emotions expressed by Filipinos through Tweets (Pippin et. al., 2015)**
 - Naive Bayes Classification on tweets : happiness, sadness, anger, fright, surprised, disgust, neutral.
 - Moderate classification performance (70% accuracy) in detecting happy and sad emotions
 - Inconclusive results on other emotions due to lack of sample phrases
 - Recommendations:
 - Enrichment of training data to represent all emotions
 - Exhaustive collection of emoticons, chat terms, jeje-mon words
-

Objectives

1. To **build a corpus of Filipino statements** reflecting these complex emotions that can be used as a baseline for further affective analyses through NLP.
2. To **train and evaluate** the performance of a **large language model for classifying Filipino statements** into seven **complex affective states**: sadness, joy, love, anger, fear, anxiety, stress



Dataset Description

Dataset for Emotion Classification

Dataset 1: Pandey (2021): [Emotion Dataset for Emotion Recognition Tasks](#)

Dataset 2: Balatska (2024): [Mental Health Sentiment Analysis](#)

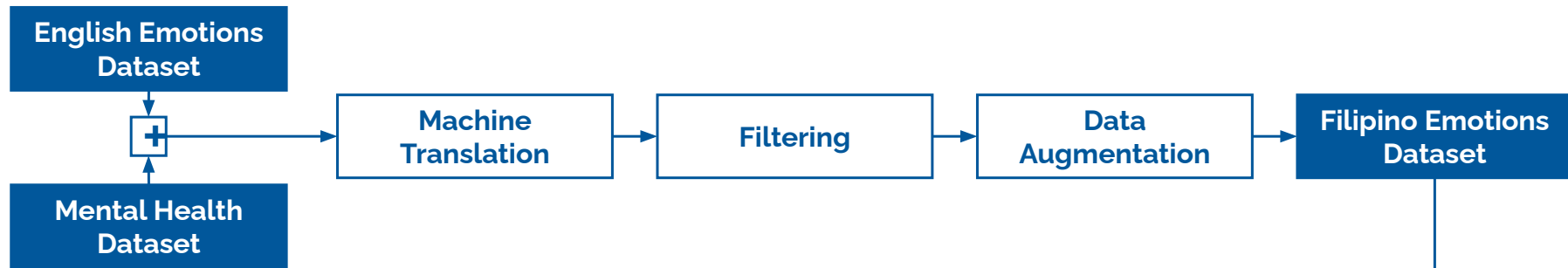
Total English Phrases : 59,210

0 - Sadness	(20,070 samples)
1 - Joy	(5,362 samples)
2 - Love	(1,304 samples)
3- Anger	(2,159 samples)
4 - Fear	(2,509 samples)
5 - Anxiety	(3,841 samples)
6 - Stress	(2,587 samples)

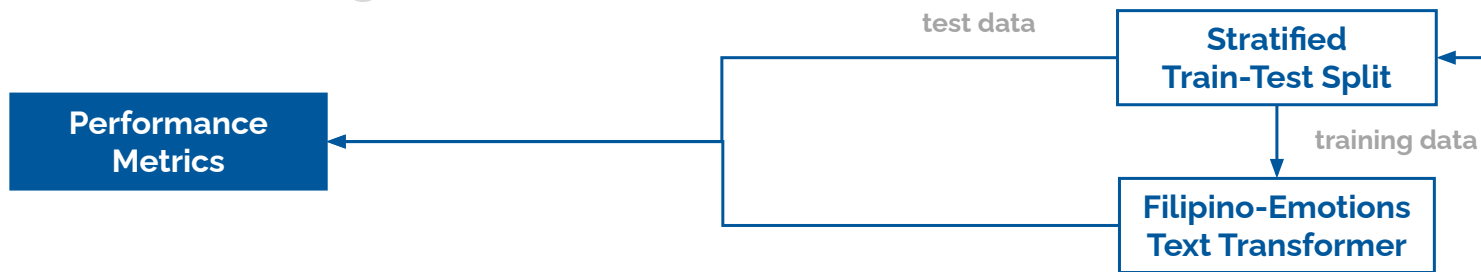
Methodology

Methodology

Data Preparation



Model Training



Data Preparation and Corpus Generation

Step 1: Machine Translation

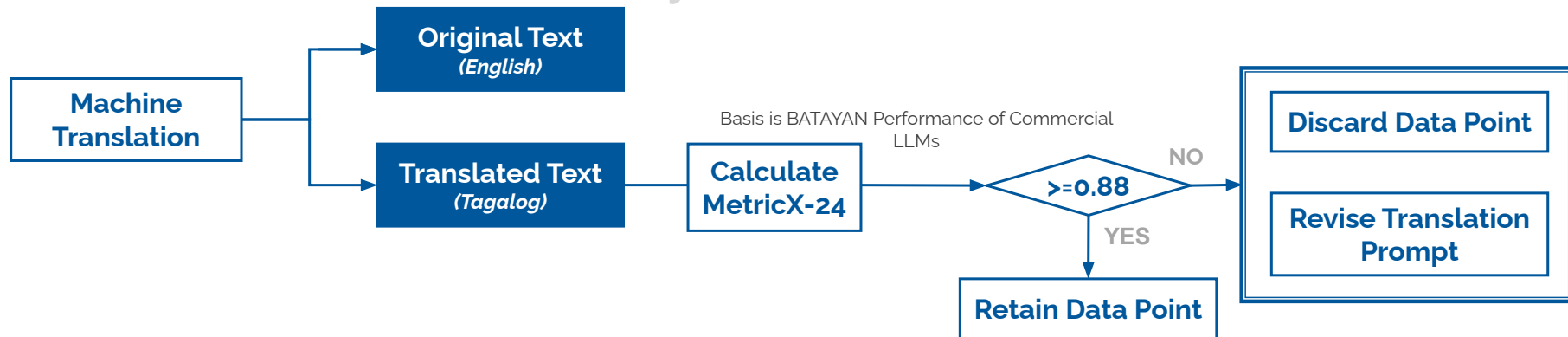


```
def build_prompt(text):  
    return (  
        "You are a translation engine. "  
        "Translate the text into natural and standard Filipino. "  
        "Maintain the original meaning, tone, and emotional content. "  
        "Do not explain, summarize, rewrite, or add any commentary. "  
        "Output only the translated sentence.\n\n"  
        f"Text: {text}\n"  
        "Translation:"  
    )
```

Data Preparation and Corpus Generation

Step 2: Evaluation and Filtering

Filter For Translation Accuracy



NOTE: There is no current [automatic] metric for evaluating Translationese in machine translation and existing studies do not involve English-Tagalog pairings (Tumaliuan et al.. 2024)

Data Preparation and Corpus Generation

Step 2: Evaluation and Filtering

Models Used:

- MetricX-24 [Batayan]
- Metric-24-hybrid-xl-v2p6 [Google]

Interpretation:

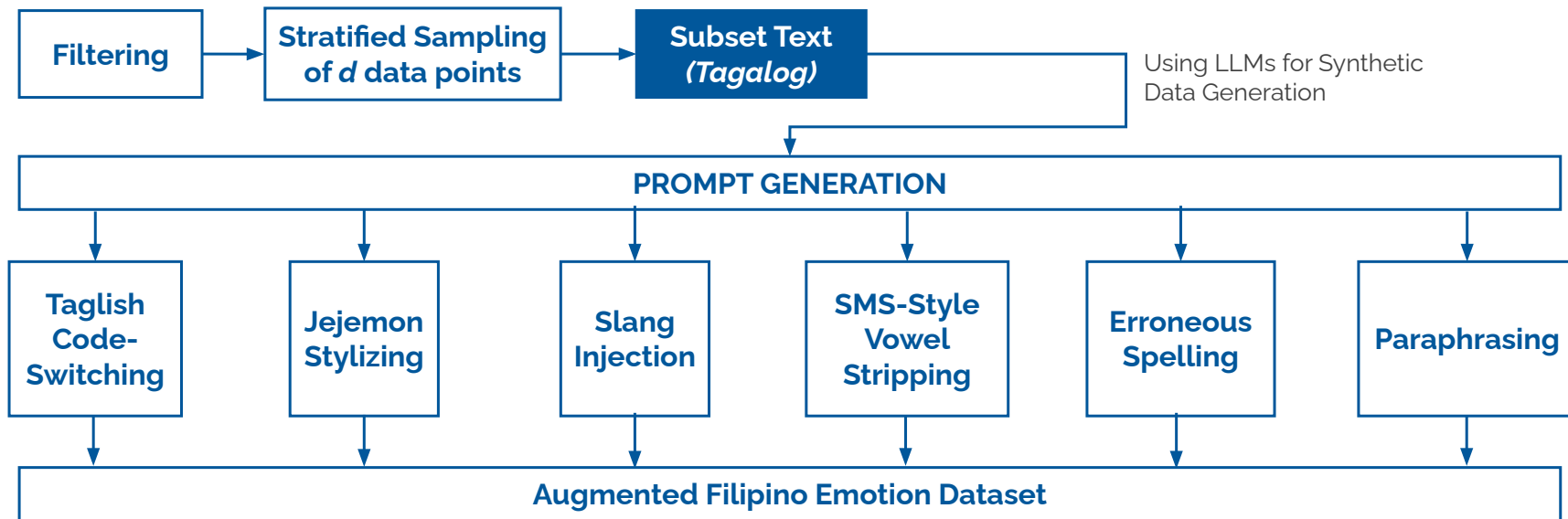
- Values are from 0 to 25: Lower is better
- Values are from 0-100 : Higher is better

Filtered: All translations with Scaled_Batayan Score > 87.24

	text	label	label_desc	translation	MetricX24_Raw	MetricX24_Scaled_BATAYAN
0	i didnt feel humiliated	0	sadness	Hindi ako nadamaang nahihiya.	0.277646	98.889417
1	i can go from feeling so hopeless to so damned...	0	sadness	Maaari akong lumipat mula sa pagiging lubos na...	0.417274	98.330905
2	im grabbing a minute to post i feel greedy wrong	3	anger	Kukuha ako ng isang minuto para mag-post, para...	0.410044	98.359825
3	i am ever feeling nostalgic about the fireplac...	2	love	Lagi akong nakakaramdam ng pag-nostalya tungko...	0.313250	98.747002
4	i am feeling grouchy	3	anger	Masama ang pakiramdam ko.	0.476642	98.093433

Data Preparation and Corpus Generation

Step 3: Augmentation



Data Preparation and Corpus Generation

Step 3: Augmentation

Filipino Translated Phrases are augmented with : **+Tag-lish**, **+Jejemon**, **+Slang**, **+Vowel Stripping**, **+Erroneous Spelling**, **+Paraphrasing**

```
# Define a Data Augmentation Prompt

def build_prompt(text):
    return (
        "You are a synthttic data generator. "
        "Transform the Filipino sentence by adding Tag-lish or natural code switching."
        "Maintain already existing code switching, if there is."
        "For example if the original sentence is: Hindi masyado kauna-unawa ang mga nakaraang "
        "pinaggagawang panghihinakit sayo, it becomes: Mahirap i-understand ung mga ginawang trauma sayo before."
        "Maintain the original meaning, tone, and emotional content. "
        "Do NOT explain, summarize, rewrite, or add any commentary. "
        "Return only the revised sentence.\n\n"
        f"Text: {text}\n"
        "Augmented:"
    )
```

```
# Jejemon Stylizing Prompt Function

def build_prompt(text):
    return (
        # Persona (Do not Remove)
        "You are a synthetic data generator specializing in Jejemon-style transformations. "

        # Indicate what kind of Data Augmentation they will do
        "Transform the Filipino sentence by converting it into Jejemon-style writing (random cap

        # Additional rules
        "Maintain readability but apply strong Jejemon traits. Maintain any existing meaning and

        # One-Shot Example
        "For example, 'Bakit ka umiiyak?' becomes: 'bhAkItt kA uMiiIyYakxX?' "

        # Prompt to minimize information loss (Do not Remove)
        "Do NOT change the underlying meaning of the text. Only stylize the spelling. "

        # Prompt to only output the needed data (Do not Remove)
        "Do NOT explain, summarize, or add commentary. Output only the revised sentence.\n\n"

        f"Text: {text}\n"
        "Augmented:"
    )
```

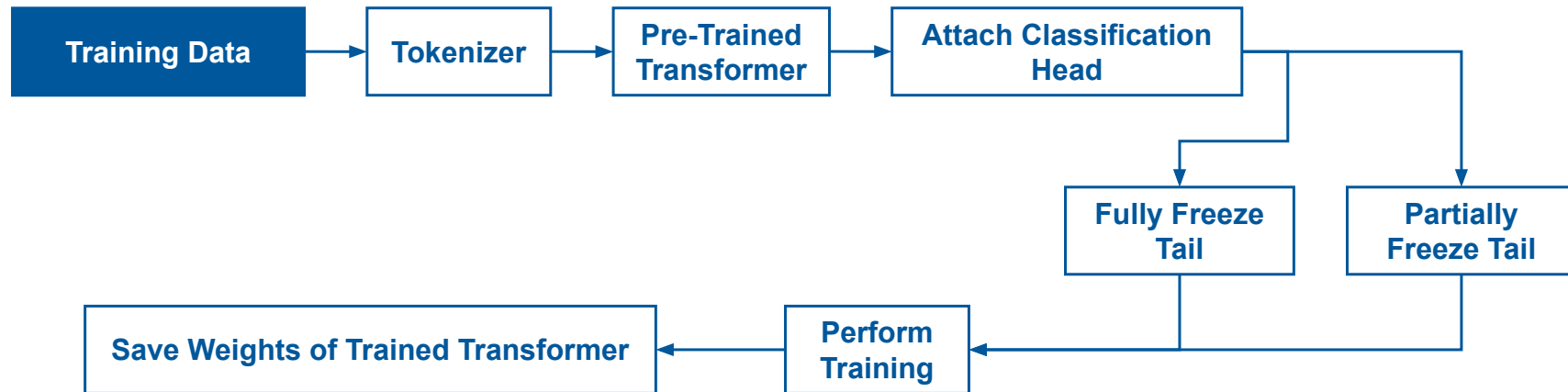

Augmented Filipino Emotion Dataset

ID	Original	Translated	Emotion	Flag
1	Sentence1	Translated1	Emotion1	Flag1
2	Sentence2	Translated2	Emotion2	Flag2
...	...	...	...	...
N	SentenceN	TranslatedN	EmotionN	FlagN

flags for post evaluation of synthetic
data augmentation impact

1. Translated Only
 2. SDA – Tag-lish
 3. SDA – Jejemon
 4. SDA – Slang
 5. SDA – Vowel Stripping
 6. SDA – Erroneous Spelling
 7. SDA – Paraphrasing
-

MODEL TRAINING



MODEL TRAINING

LLM Base Model Architecture

Base Model	Description	Hidden Size	Hidden Layers	Attention Heads	Vocabulary Size
jcblaise/roberta-tagalog-base	RoBERTA trained on Tagalog	786	12	12	30,000
jcblaise/roberta-tagalog-large	(larger version)	1024	24	16	30,000
xlm-roberta-base	Multilingual trained on CommonCrawl data	786	12	12	250,002
bert-base-multilingual-cased	Multilingual trained on Wikipedia	786	12	12	119,547

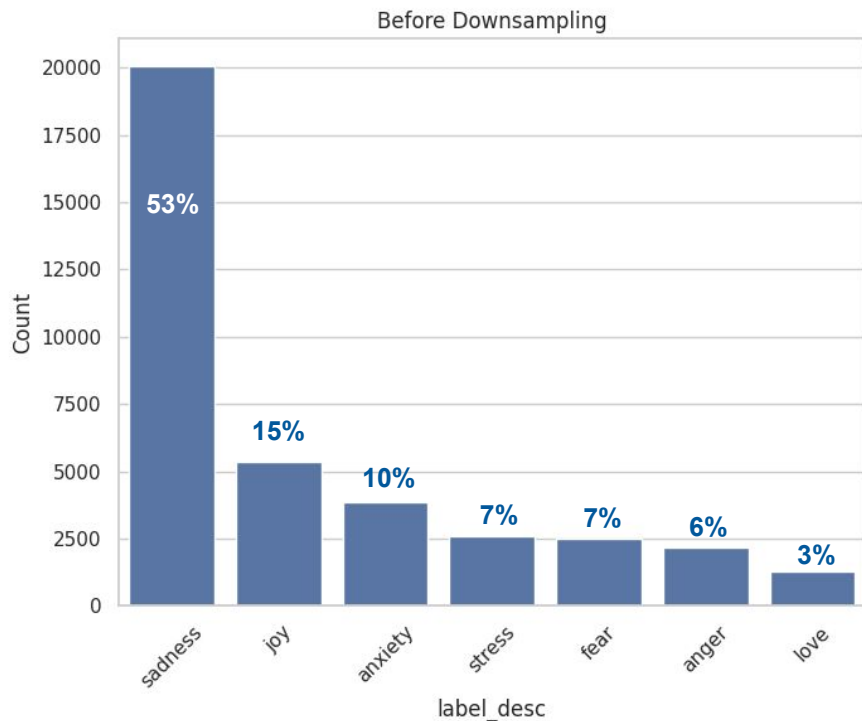
Results & Analysis

Results

(Part 1: Filipino Emotions Corpus Generation)

1. To build a corpus of Filipino statements reflecting these complex emotions that can be used as a baseline for further affective analyses through NLP.
-

Original Dataset

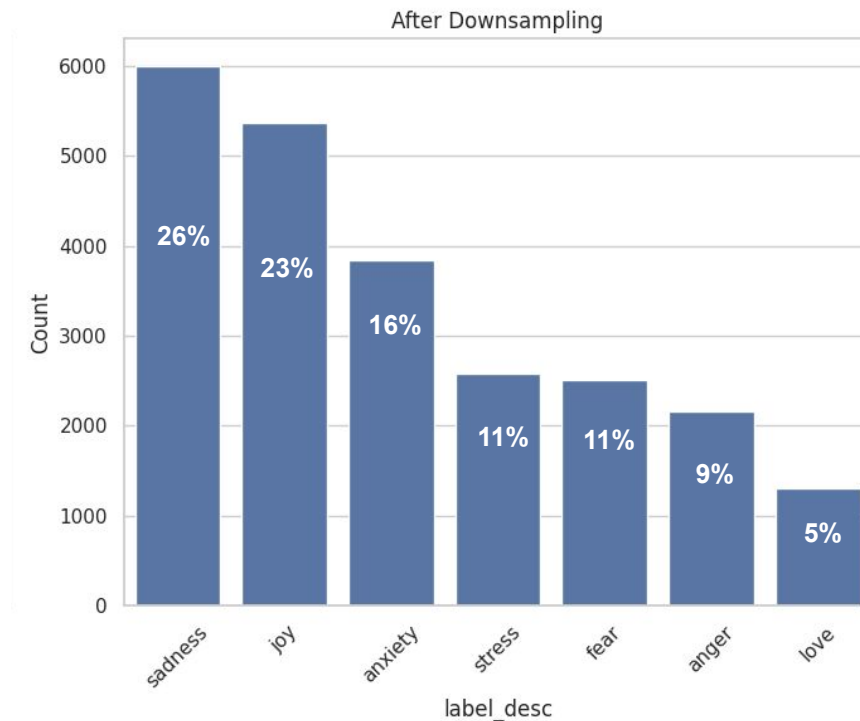


Original Samples: 59k+
(no augmentation)

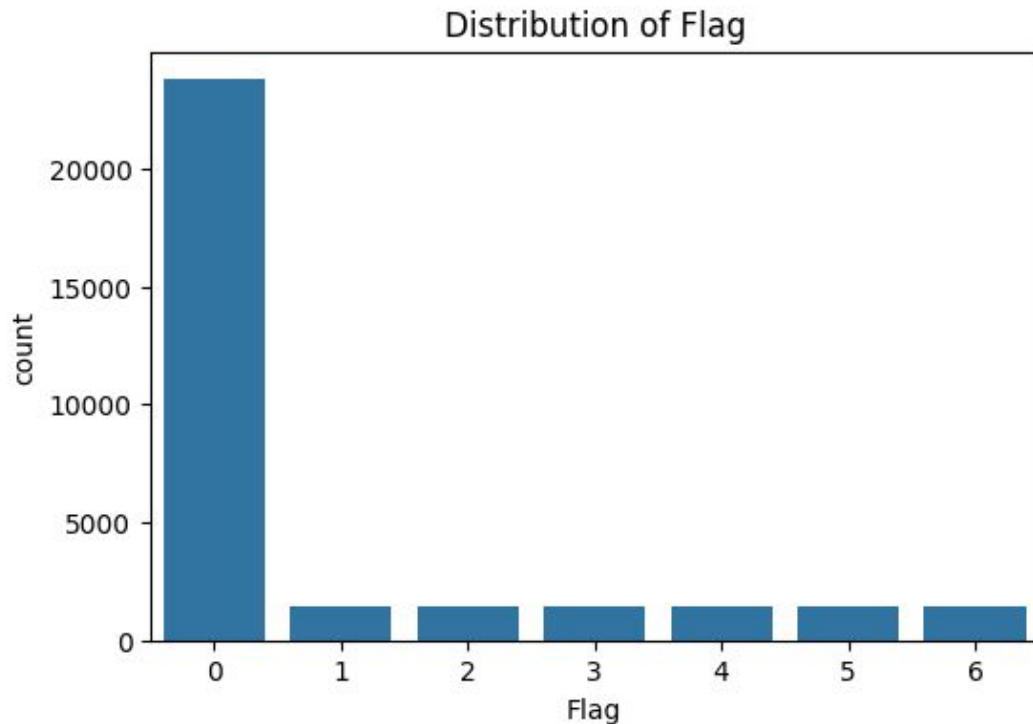
Generated Synthetic Filipino Emotions Dataset

After randomly retaining 6000 samples
from the most dominant class (sadness)

-> 23762 samples



Distribution of Augmentations



flags for post evaluation of synthetic data augmentation impact

0 - Translated Only

1 - [SDA] – Tag-lish

2 - [SDA] – Jejemon

3 - [SDA] – Slang

4 - [SDA] – Vowel Stripping

5 - [SDA] – Erroneous Spelling

6 - [SDA] – Paraphrasing

Total Phrases: 32162 phrases

Distribution after Data Augmentation (1400 per augmentation)

Sample Data Augmentations

Sample of Taglish

im under a lot of stress and feeling overwhelmed	Grabe ang stress ko, parang lumulubog na ako.	fear
i hold the bow it make me feel cool	Hawak ko 'tong pana, parang nakakakilig ah.	joy
i feel can be really popular in the underground if they get themselves out there and thank god for this i m looking at you toby and tunji	Nararamdaman kong baka super sikat sila sa ilalim kung lalabas sila, salamat na lang sa Diyos dahil dito, nakatingin ako sa inyo, Toby at Tunji.	joy

Sample of Jejemon

i feel funny about mothers day	nAkAkAkAbA sA AkIn Ang m0Th3r's dAyyyy.	fear
i feel there is a shortage of loyal people whom you can trust	nArArAmDaMaN kOng kuLanG aNg mAgA tApAt na mApagkAtlwAlaAn.	love
i feel shaken by it and im far far above the age group targeted	Kinakabahan ako dito, eh, parang layo na ako sa target age.	fear

Sample Data Augmentations

Sample of Slang

i feel weird	Nararamdaman kong kakaiba, legit.	fear
i feel like beloved	Nararamdaman kong mahal na mahal, legit amp.	love
i feel like beloved	Nararamdaman kong mahal na mahal, legit amp.	love

Sample of Vowel Stripping

i feel jealous with them why they can	niinggit ako s knla, bkit kaya sln?	anger
i feel like i shouldnt have even bothered	prng hndi k dpat pnagalaban.	anger
i feel there is really no point in me loving him after getting to know his true color	nramdaman kng wla ng sbi ang pgmhal ko sknya mtp lng ang trng klr nya.	love

Sample Data Augmentations

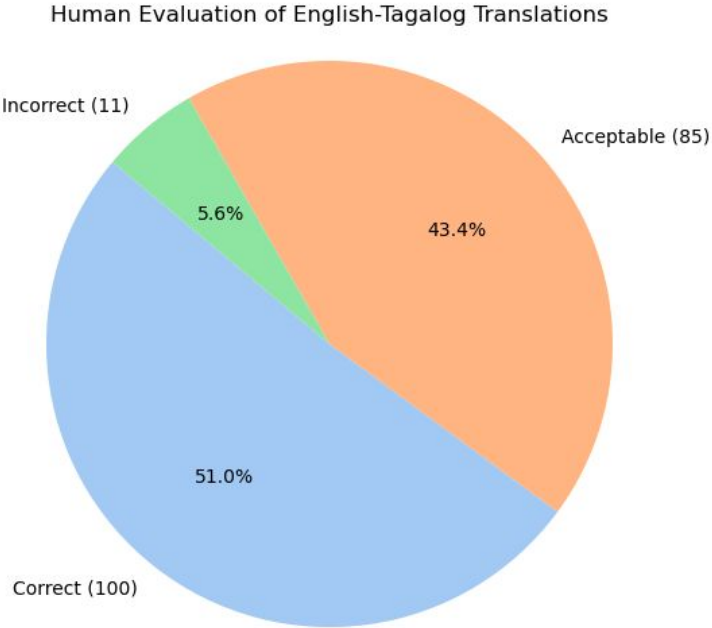
Sample of Erroneous Spelling

i feel outraged about this type of thing	Nagaglit ako sa ganitong uri ng byahe.	anger
i feel cold few days	Naraamdaman kong malmig ang mga nakaraang araw.	anger
i could feel my feet getting agitated once i got to the metropolitan pavilion	Nararamdaman kong nag-aagaw ang aking mga paa nang makaringt ako sa Metropolitan Pavilion.	anger

Sample of Paraphrasing

i don t feel special and when i feel alone in this busy ever moving world	Parang hindi ako nabibigyan ng pansin, kahit na palagi akong nakapaligid ng tao. Nakakaramdam ako ng sobrang pag-iisa, eh.	joy
i would cry scream kick at the door and feel terrified	Umiiyak, sigaw na sigaw, tatapakan ko 'yung pinto, at sobrang takot ako.	fear

Human Evaluation of English-Tagalog Translations



Examples of Incorrect English-Tagalog Translations

English	Tagalog
i find myself feeling irritable or depleted i run through a mental checklist have i worked out	Kapag nararamdaman kong nagiging maasim o pagod ang aking pakiramdam, sinusuri ko sa aking isipan kung nag-ehersisyo na ba ako.
i feel rather agitated by our sliding door that keeps getting stuck	Grabe, ang sakit na sa ulo ko sa pinto na palagi kong binubuksan para sa mga tao.
I swear anj from td why sii, I like to be nervous tbtb for no reason ¯_¯ ¯_¯ ¯_¯ ¯_¯	Pinapangako ko, ang jowa ko mula sa TD, bakit siya, gusto kong maging kinakabahan nang walang dahilan.
i feel outraged about this type of thing	Nagaglit ako sa ganitong uri ng byahe.
i am walking around feeling quite tortured because i spent so many hours on it and it is still not finished but i have learned a few things	Nakakapagod talaga maglakad ng ganito katagal, kahit maraming oras na ako dito, hindi pa rin tapos. Pero oo, natutunan naman ako ng ilang bagay.

Results

(Part 2: Classification Model)

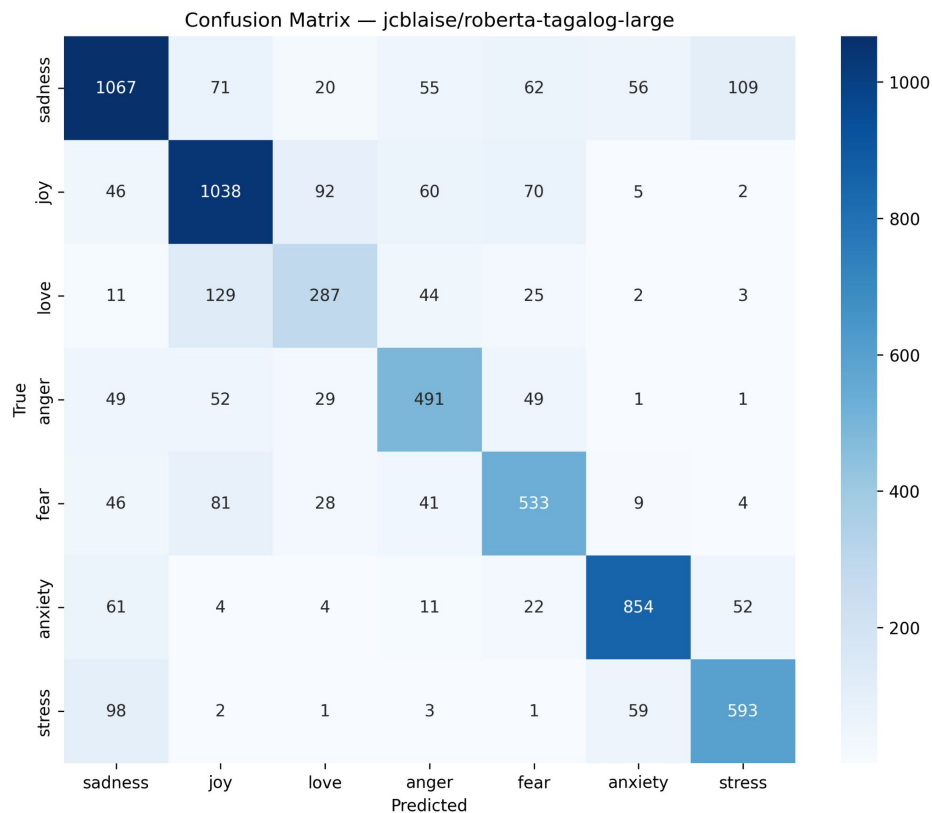
2. To train and evaluate the performance of a **large language model** for **classifying Filipino statements** into seven **complex affective states**: **sadness, joy, love, anger, fear, anxiety, stress**

Fully Frozen Tails

Base Model	Accuracy	Precision	Recall	F1-Score
jcblaise/roberta-tagalog-base	0.5514	0.5205	0.5030	0.5045
jcblaise/roberta-tagalog-large	0.5818	0.5582	0.5341	0.5348
xlm-roberta-base	0.4313	0.3702	0.3351	0.3004
bert-base-multilingual-cased	0.4359	0.3931	0.3436	0.3164

Partially Frozen Tails

Base Model	Accuracy	Precision	Recall	F1-Score
jcblaise/roberta-tagalog-base	0.7289	0.7120	0.7056	0.7078
jcblaise/roberta-tagalog-large	0.7559	0.7412	0.7406	0.7406
xlm-roberta-base	0.6328	0.6036	0.5919	0.5946
bert-base-multilingual-cased	0.5657	0.5431	0.4983	0.4924



Best Performing Model - Roberta Tagalog Large (Partially Fine Tuned)

Accuracy	0.7559
Precision	0.7412
Recall	0.7406
F1-Score	0.7406

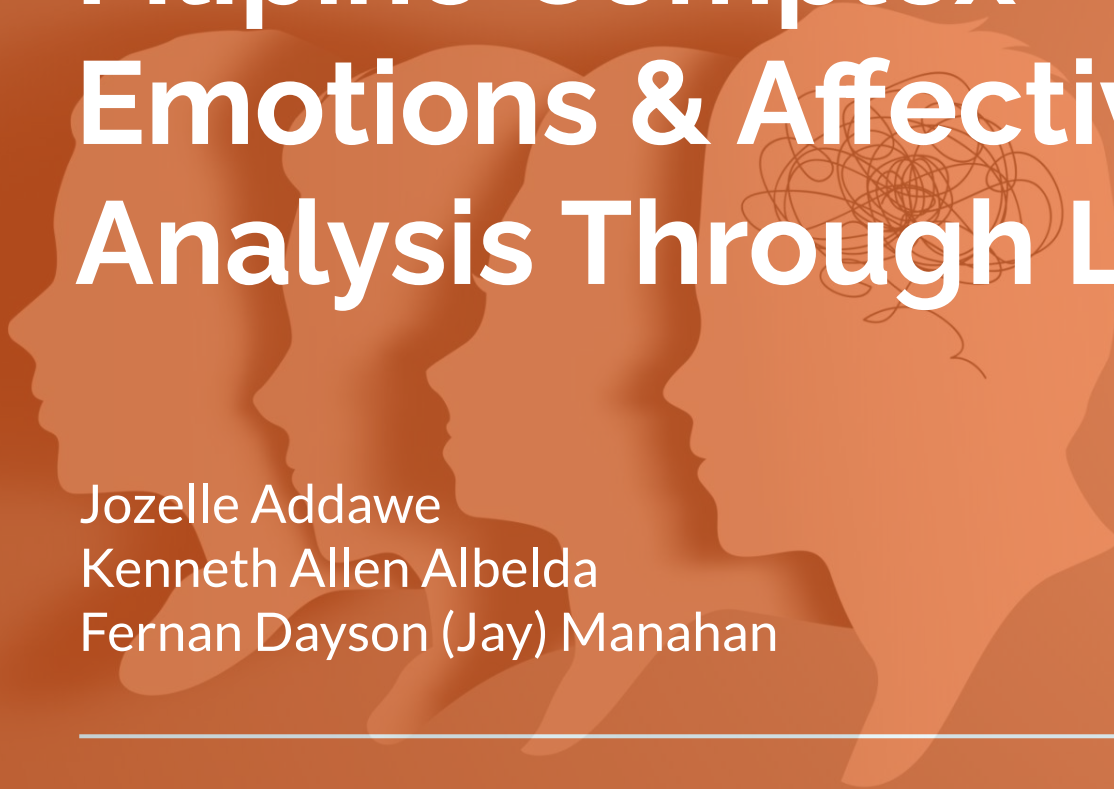
Summary

- This experiment presented a framework for classifying Filipino text among 6 emotions by utilizing a labelled English emotion dataset and machine translation.
 - LLM used is capable of Machine Translation of English to Filipino at ~94.4% success rate (based on Human Evaluation).
 - Pretrained models are tested for classifying the translated emotions dataset, with the best performing model being the Roberta-Tagalog-Largemodel by jcbraise (Cruz & Cheng, 2021) achieving 75% accuracy.
 - The generated filipino emotion dataset of 32,162 Filipino statements—including Filipino-specific morphological and semantic nuances of Taglish, jejemon, slang, and vowel-stripping—may be used for further affective analyses.
-

Issues Encountered

1. Not all generated texts have been fully manually graded/evaluated due to time constraints
 2. Character limit for text translation truncated some of the filipino texts for translation. (we have samples having more than the character limit)
 3. GPU availability
 4. More data on the undersampled cases
-

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